



6G4Z3006 Calculus: Mock exam 01

Duration : 3 hours

Instructions to students

- The 2025/26 exam will be a 3 hour exam.
- In the 2025/26 exam you will be given **two compulsory questions** with questions drawn from all parts of the module and then a choice of any **two** questions from a set of four optional questions (one question each based on the topics of sequences, series, differentiation and integration). Each of the four questions you will answer will be worth 25 marks.

Permitted materials

- Faculty formula handbook
- Students are permitted to use their own calculators without mobile communication facilities.

Section A – answer both questions

1. (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Use the concept of limiting value of a function to give the definition of the *derivative* of f at the point $a \in \mathbb{R}$. [3]
- (b) Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ and a point $a \in \mathbb{R}$ in its domain such that f is *not* differentiable at a . Hint: you could define f as a certain piecewise-linear function. [3]
- (c) State and prove the *sum rule* of differentiation. You can use without proof any relevant properties of the limiting values of functions, providing you clearly state these in your proof. [8]
- (d) State the *chain rule* of differentiation. [3]
- (e) State the *quotient rule* of differentiation. [3]
- (f) Give a proof of the quotient rule by considering the quotient u/v , of functions u and v , in the form [5]

$$u/v = u \cdot (r \circ v),$$

i.e. as the product of u with the composition r after v , where r is the reciprocal function defined by

$$r(x) = \frac{1}{x}.$$

You can use without proof the results of the product and chain rules, as well as the known derivative of the reciprocal function, i.e.

$$r'(x) = \frac{-1}{x^2}.$$

Section A – answer both questions

2. (a) State both parts of the *Fundamental Theorem of Calculus*. Suppose the function $F(x)$ is defined by [8]

$$F(x) = \int_0^x (t^2 + \cos(t)) dt.$$

Use the Fundamental Theorem of Calculus to find $F'(\pi/2)$, i.e. the derivative of F at $\pi/2$.

- (b) Suppose that $\{x_n\}_{n=1}^{\infty}$ is a sequence of real numbers. Give the (ϵ, N) -definition for the convergence of the sequence $\{x_n\}_{n=1}^{\infty}$. [3]

- (c) Suppose that $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are two sequences of real numbers, and that $\alpha, \beta \in \mathbb{R}$. Use the definition from the previous part to prove that if $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are two convergent sequences with $a_n \rightarrow a$ and $b_n \rightarrow b$ as $n \rightarrow \infty$ then the linear combination sequence $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}$ is also convergent and [6]

$$\alpha a_n + \beta b_n \rightarrow \alpha a + \beta b, \text{ as } n \rightarrow \infty.$$

- (d) Use concepts of sequence convergence and partial sums to give the definition for the convergence of an infinite series $\sum_{n=1}^{\infty} a_n$. [3]

- (e) Suppose that $\alpha, \beta \in \mathbb{R}$. Use parts (c) and (d) and relevant properties of sequence convergence to prove that if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are two convergent series of real numbers with [5]

$$\sum_{n=1}^{\infty} a_n = a \text{ and } \sum_{n=1}^{\infty} b_n = b,$$

then the linear combination series $\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n)$ is convergent also and

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b.$$

End of Section A

Section B – answer any two questions

3. (a) Use the limit definition of the derivative to obtain the derivative of the function f defined by the formula [5]

$$f(x) = x^6.$$

Point out clearly where you make use of any properties of the limiting values of functions.

- (b) Consider the function g defined by the formula [6]

$$g(x) = x^4 \sin(2x).$$

Use the linearity, product and chain rules to obtain the first and second order derivatives, $g'(x)$ and $g''(x)$, of g . Point out clearly where you are using each of the rules.

- (c) Use the concepts of the *differrentiation* and *stationary point* to find the rectangle of maximal area that can be drawn inside the unit circle. [10]
- (d) Let u, v, w, z be four differentiable functions of the variable x . Use a combination of the product rule and quotient rule to fully expand the derivative [4]

$$\frac{d}{dx} \left(\frac{uv}{wz} \right),$$

and show how it depends on the individual functions u, v, w, z and their derivatives.

Section B – answer any two questions

4. (a) Evaluate the following integrals. You can use various integration techniques and/or trigonometric formula but you must explain what you are doing and clearly reference any standard integrals you make use of.

(i)

[6]

$$I_1 = \int_0^{\pi/2} x^2 \sin(x) dx$$

- (ii) Use the substitution $u = \sqrt{x+2}$ to evaluate the integral

[6]

$$I_2 = \int_{-2}^2 \frac{\sqrt{x+2}}{x+6} dx$$

- (b) The Riemann integral of a function $f(x)$ between the limits $x = a$ and $x = b$ is defined as

[13]

$$\int_a^b f(x) dx = \lim_{\substack{n \rightarrow \infty \\ \Delta x \rightarrow 0}} \left(\sum_{i=1}^n f(x_i^*) \Delta x_{i-1} \right),$$

where the x_i ($0 \leq i \leq n$) form a subdivision of the interval $[a, b]$ with

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b,$$

$\Delta x_{i-1} = x_i - x_{i-1}$ is the width of the i^{th} subinterval,

$\Delta x = \max\{\Delta x_i : 0 \leq i \leq n-1\}$ and x_i^* is a representative point from the i^{th} subinterval, i.e. $x_{i-1} \leq x_i^* \leq x_i$.

In this question you will evaluate the Riemann integral for the function $f(x) = x^2$ and the limits $a = 3$ and $b = 5$. You should use the equally spaced interval

$$x_i = 3 + i \frac{2}{n}, \quad (i = 0, \dots, n),$$

and the start point of each subinterval as the representative point, i.e.

$$x_i^* = 3 + (i-1) \frac{2}{n}.$$

Set up and evaluate the limit in the Riemann integral, using known summation formulae, to show that

$$\int_3^5 x^2 dx = 98/3.$$

You can use, without proof, any of the following summation formulae

$$\sum_{i=1}^n i = \frac{n(n+1)}{2},$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Section B – answer any two questions

5. (a) Let the sequence $\{b_m\}_{m=2}^{\infty}$ be defined by the formula [5]

$$b_m = \frac{3m - 5}{m}.$$

By considering the ratio or difference of its consecutive terms, prove that this sequence is increasing for all $m \geq 2$.

- (b) Let the sequence $\{c_m\}_{m=1}^{\infty}$ be defined by the formula [7]

$$c_m = \frac{3m^3 - 2m + 1}{-4m^3 + 2}.$$

Show how the algebra of limits theorem for convergent sequences can be used to prove that $\{c_m\}$ is a convergent sequence and to determine its limit. Make it clear in your answer where you are using each part of the theorem that you rely on.

- (c) When the Newton-Raphson method for root finding is applied to the equation

$$f(x) = x^2 - 5 = 0,$$

it produces a sequence, $\{x_n\}_{n=0}^{\infty}$, of rational estimates of $\sqrt{5}$, beginning with the number $x_0 = 3$ and for $n \geq 0$ the subsequent estimates are given by the recurrence relation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 5}{2x_n}.$$

In this question you will construct a proof that justifies the claim of the Newton-Raphson method, i.e. that the sequence does indeed converge to the limit $\sqrt{5}$.

- (i) Use the recurrence relation above to prove, by induction, that for all $n \geq 0$ the sequence elements satisfy $x_n > \sqrt{5}$. [6]
- (ii) Use the result of part (i) to prove that the sequence $\{x_n\}_{n=0}^{\infty}$ is a decreasing sequence. [2]
- (iii) Briefly explain how the Monotone Convergence Theorem can be used to prove that $\{x_n\}_{n=0}^{\infty}$ is a convergent sequence. [2]
- (iv) Finally, use the algebra of limits theorem to prove that $x_n \rightarrow \sqrt{5}$ as $n \rightarrow \infty$. [3]

Section B – answer any two questions

6. (a) Consider the infinite series

[8]

$$\sum_{n=1}^{\infty} \frac{2 + 3^n}{5^{n+2}}.$$

Use the result about the convergence of geometric series to prove that the series defined above is convergent and determine its sum. You will also need to make use of the theorem about linear combinations of convergent series.

- (b) (i) Use a suitable comparison test to prove that the series

[5]

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2 + n^2},$$

is a convergent series. You can make use of the known convergence of any hyper-harmonic series.

- (ii) Use the ratio test to prove that the series

[5]

$$\sum_{m=1}^{\infty} \frac{3^m}{(m+2)!},$$

is a convergent series. *Note that ! denotes the factorial operation.*

- (c) Obtain the partial fraction expansion of

[7]

$$\frac{15}{m^2 + 5m + 4},$$

and use it to obtain the sum of the infinite series

$$\sum_{m=1}^{\infty} \frac{15}{m^2 + 5m + 4},$$

by considering cancellation effects in the partial sums of this series.

End of Section B
End OF QUESTIONS